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Coffee Leaf Disease Classification Report

Report

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1. Abstract

In this report, two image classification models trained on the same coffee leaf disease dataset were compared. The objective of this study was to classify images of coffee leaves into one of three classes – healthy leaves, coffee leaf rust, and Phoma disease. The models tested in this experiment were a MobileNetV2 transfer learning model and a Simple CNN. Both models used the same training, validation, and testing folders, the same batch size, and input images of 224 x 224 pixels. The three output classes for both models were the same as well. The Simple CNN model performed better in terms of test accuracy, achieving 95.88%, compared to 94.43% in the case of the MobileNetV2 model. However, in terms of the training time, the Simple CNN model needed 82 minutes for training while MobileNetV2 needed only 16 minutes for the frozen-base model run.

2. Introduction

Coffee leaf diseases may decrease crop productivity and quality. Machine vision-based computer models can assist farmers with automated disease detection by learning from labeled images of healthy and diseased leaves. In the current project, the task was not only to build a model that performed classification accurately, but also to compare two methods and analyze the differences between the chosen models. Such analysis was useful since the model with the best accuracy was not necessarily the most efficient one to deploy. In this paper, the criteria for comparison included such parameters as accuracy, loss, performance per class, the number of model parameters, preprocessing, limitations and possible problems with the dataset.

Two models discussed in this experiment belonged to two widely-used types of image classification techniques. The first one used transfer learning based on pre-trained MobileNetV2. Only a classification head was trained while the remaining feature extraction part of the model was kept frozen. The second approach used a custom CNN architecture which learned convolutional filters during the training process using the dataset. Both of these models were applied depending on the conditions: transfer learning in case there was a lack of labeled data and vice versa.

2.1. Related Work and Scientific Context

Previous works demonstrate that accurate plant disease image classification could be accomplished if both training and validation images come from the same data set; however, it is not always the case. Mohanty et al. trained their deep learning algorithm on the PlantVillage data set and obtained 99.35% classification accuracy. Nonetheless, when their model was trained using other online images, the overall accuracy reached just 31.4%. It follows that although high accuracy within the same data set demonstrates feasibility, it still should not be considered a sufficient condition for deployment [8].

This applies even for more specialized models designed to recognize coffee-related diseases. Esgario et al. used CNN-based classifiers to distinguish different coffee biotic stresses and their severity.

The authors obtained 95.24% accuracy for classification and 86.51% for estimation of coffee leaves' disease severity [9]. In this paper, the authors also noted that some errors occurred due to the resemblance of the disease's signs and due to imbalanced dataset. Consequently, failure cases in scientific studies provide information about the limitations of the approach and necessary improvements.

Several specific coffee leaf data sets (e.g., RoCoLe and JMuBEN/JMuBEN2) have been developed recently to improve availability of such images. For example, RoCoLe data set includes images of Robusta coffee leaves that are either healthy or contain symptoms of various diseases. Another example is JMuBEN and JMuBEN2 which include 58,555 images of Arabica coffee leaves (disease and healthy images) [10, 11].

3. Dataset Description

The external dataset used in this experiment was "A Machine Learning Dataset for Classification of Common Coffee Leaf Diseases in Uganda." The dataset was posted on Mendeley Data with DOI: 10.17632/k36wnd6knb.1 [1]. According to the source page, the dataset contained a set of labeled images of coffee leaf diseases collected under various lighting conditions on farms in Uganda. These images were classified into three classes: healthy leaves, leaf rust, and Phoma disease [1]. Images were stored as JPEG files of 256x256 pixels, whereas the experiment loaded the images at 224x224 pixels prior to the training [1].

An important point about reproducibility of this dataset was noted. According to the description, the Mendeley source provided 3,312 images, but the local dataset contained a slightly different number of files before cleanup. Corrupted files were then cleaned, and 2,252 training images, 483 validation images, and 485 test images were used for the model training in total of 3,220 images. In this report, the external source was used for documenting the dataset origin, and model results were obtained with the exact numbers provided by the experimental outputs.

Table 1: Count of usable images after image cleansing.

Class	Training images	Validation images	Test images	Total usable images
Healthy leaves	754	163	162	1,079
Leaf rust	721	154	156	1,031
Phoma	777	166	167	1,110
Total	2,252	483	485	3,220

3.1. Dataset Balance and Discrepancy Check

The source claims that there were 3,312 images in total, however, our local experimentally cleaned dataset included only 3,220 images usable for training. This can be attributed to the image cleaning process we did for the local experiment, during which unreadable and/or corrupted images were deleted. Therefore, our results must be interpreted as the results on the cleaned local dataset, and not on the whole source dataset. To improve the reproducibility of the work, the final work should state the number of files that were excluded, as well as have an exact manifest of the exact images used for the train/validation/test sets.

It is worth noting that the final three-class dataset was relatively balanced: the biggest class consisted of Phoma with 1,110 images, whereas the smallest class consisted of leaf rust with 1,031 images. This means that there was little class imbalance, with the imbalance factor equal to around 1. As such, there probably isn't much to be said regarding the difference between MobileNetV2 and Simple CNN in regard to class imbalance. Moreover, the majority of images were not healthy leaves even when taking diseased classes into account: healthy leaves accounted for 1,079 images, while diseased leaves had 2,141 images. Nevertheless, the three classes were balanced.

$$IR = \frac{n_{\max}}{n_{\min}} = \frac{1110}{1031} = 1.08$$

Table 2: Class group images and percentage.

Class group	Images	Percentage
Healthy leaves	1,079	33.5%
Leaf rust	1,031	32.0%
Phoma	1,110	34.5%
Diseased combined	2,141	66.5%

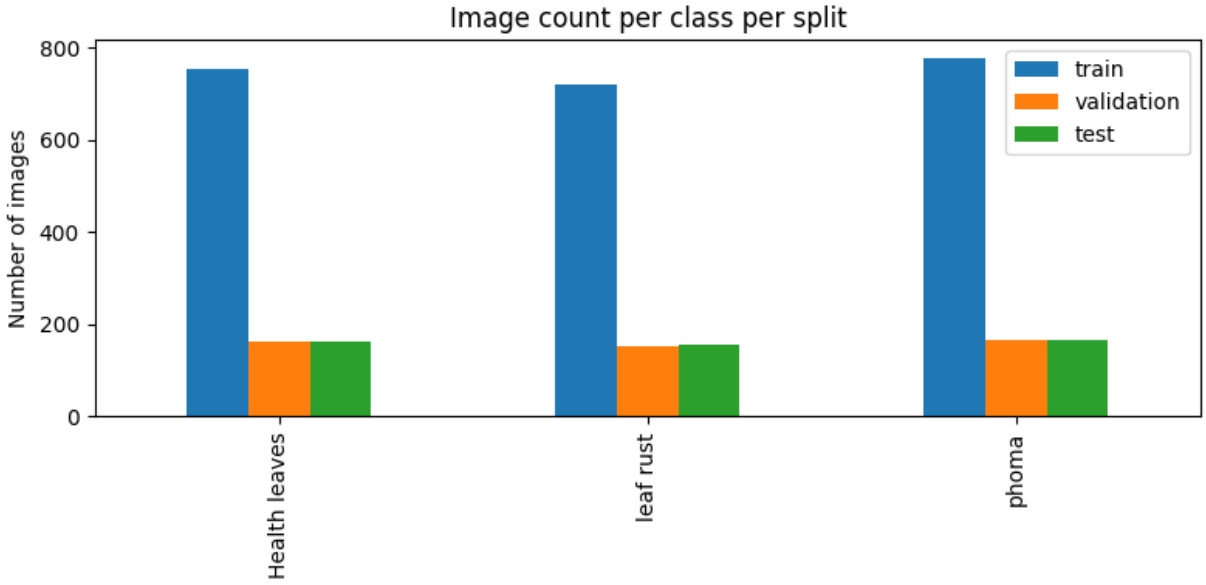


Fig. 1. Count of images per class and split after image cleansing.

4. Preprocessing and Experimental Setup

The images were preprocessed and prepared for training in the same way for both models. The images were read from folders and scaled to 224 x 224 pixels. They were batched with batch size 32 and transformed to categorical labels. The training set was augmented with random rotations, width shifts, height shifts, horizontal flips, and zooming. Validation and test data were not augmented; they were only rescaled.

Models were compiled with categorical cross-entropy as a loss function and Adam optimizer with the initial learning rate of 0.001. Early stopping, learning-rate reduction on plateau, and best-model checkpointing were used to reduce overfitting and keep the best validation model. Finally, the classification reports were generated with scikit-learn using its classification_report function [4].

A methodology limitation was related to the MobileNetV2 preprocessing pipeline. According to the MobileNetV2 documentation in Keras, rescaling to [0..1] interval was not recommended. The documentation recommended using the MobileNetV2 preprocessing function to rescale inputs to [-1..1] interval. Even though the model achieved very good results, its preprocessing pipeline was not fully aligned with the recommended transfer-learning preprocessing method.

5. Model 1: MobileNetV2 Transfer Learning

Model 1 applied MobileNetV2 for feature extraction with transfer learning. Specifically, MobileNetV2 was proposed as an efficient mobile network with inverted residuals and linear bottlenecks [5]. Similarly, the Keras application guide mentioned that MobileNetV2 applied inverted residual blocks with bottlenecks and supported ImageNet pre-trained weights [6].

The MobileNetV2 base was loaded with ImageNet weights, the original top classifier was removed, and the base was frozen. The new classification head contained global average pooling, a 0.3 rate dropout, and a dense softmax classifier with three output classes. Thus, the workflow matched the typical transfer learning pipeline as follows: loading pre-trained base model, freezing the base model, creating and stacking the new model on top, and then training the classifier on the target data [7].

Additionally, an optional fine-tuning stage was included where the top layers of MobileNetV2 were unfrozen and trained with a reduced learning rate. The results comparison included the primary MobileNetV2 model evaluation score of 94.43% test accuracy.

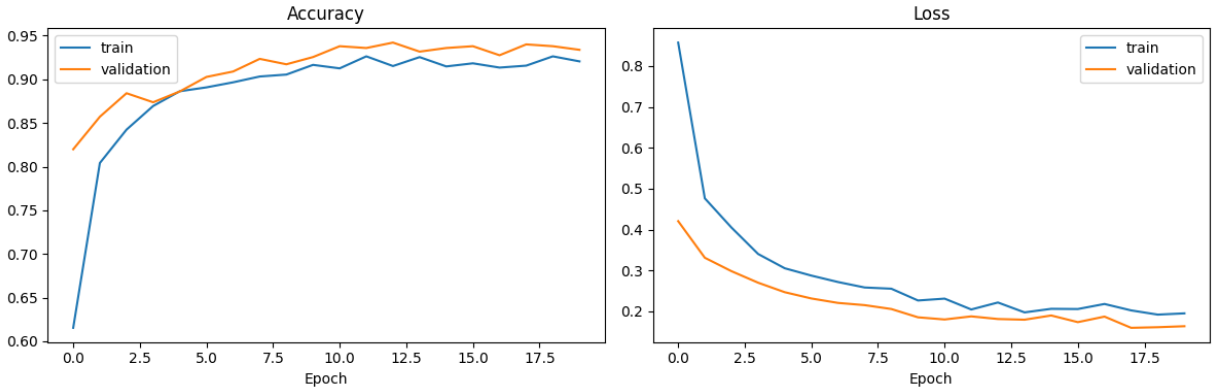


Fig. 2. Training and validation accuracy and loss curves for MobileNetV2 with frozen base model.

6. Model 2: Simple CNN

Model 2 was a custom convolutional neural network trained from scratch. In contrast to Model 1, Model 2 did not use pre-trained ImageNet weights. The architecture consisted of several convolu-

tional blocks with Conv2D layers, batch normalization, max pooling, and dropout. The last classifier contained dense layers with three class softmax outputs. Therefore, this design enabled Model 2 to extract features from the coffee leaf images directly. Still, the training time was longer compared to MobileNetV2 since all trainable filters needed to be learned from scratch using the project dataset.

Table 3: Summary of the MobileNetV2 transfer learning model and Simple CNN model.

Component	MobileNetV2 transfer learning model	Simple CNN model
Architecture type	MobileNetV2; include_top=False; ImageNet weights	Custom CNN trained from scratch
Training status	Frozen during the main training run	All model weights trained from the coffee leaf data
Layers	GlobalAveragePooling2D; Dropout; Dense layer with 3 softmax outputs	Conv2D, BatchNormalization, MaxPooling2D, Dropout; Dense layer with 3 softmax outputs
Total parameters	2,261,827	1,442,851
Trainable parameters	3,843	1,442,851
Number of epochs	20 epochs	22 epochs
Total training time	969 seconds / 16.15 minutes	4,919.10 seconds / 81.99 minutes

The training curves demonstrated gradual improvement and slightly unstable validation loss in the initial training epochs. This was expected for a CNN model trained from scratch due to randomly initialized network weights. Once the network learned useful visual filters from the input images, validation accuracy increased, and the model restored the best performance weights according to the callback settings.

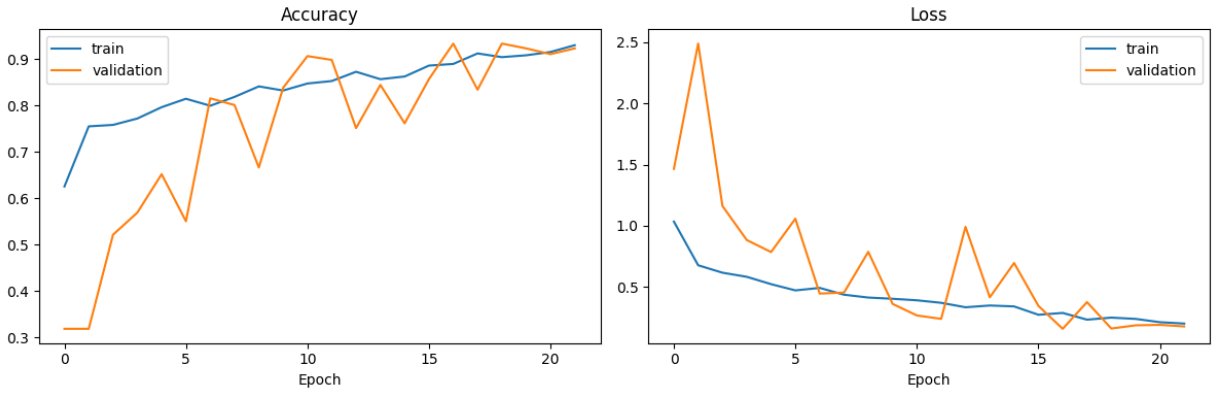


Fig. 3. Training and validation accuracy and loss curves for Simple CNN model.

7. Results

The same hold-out test set of 485 images was used for both models. This was an important criterion for comparability since both models' performance was measured against the same image distribution and class labels. The Simple CNN model yielded the highest overall test accuracy and lowest test loss value in this run.

Table 4: General comparison between both models based on the experimental outputs.

Metric	MobileNetV2 transfer learning	Simple CNN	Best result
Test accuracy	94.43%	95.88%	Simple CNN
Test loss	0.1630	0.1158	Simple CNN
Total parameters	2,261,827	1,442,851	Simple CNN smaller total model
Trainable parameters in main run	3,843	1,442,851	MobileNetV2 fewer trainable parameters
Training time	969 seconds / 16.15 minutes	4,919.10 seconds / 81.99 minutes	MobileNetV2 faster training
Training strategy	Transfer learning	From scratch	Depending on objective

Table 5: Precision, recall, and F1-score per class in the classification reports.

Class	MobileNetV2 precision	MobileNetV2 recall	MobileNetV2 F1	Simple CNN precision	Simple CNN recall	Simple CNN F1
Healthy leaves	0.95	1.00	0.97	0.99	0.98	0.98
Leaf rust	0.89	0.97	0.93	0.91	0.98	0.94
Phoma	1.00	0.87	0.93	0.98	0.92	0.95
Macro average	0.95	0.95	0.94	0.96	0.96	0.96
Weighted average	0.95	0.94	0.94	0.96	0.96	0.96

8. Model Comparison and Discussion

The Simple CNN yielded the highest numerical score in this run, achieving a test accuracy of 95.88%, compared to 94.43% by MobileNetV2. The gap between the two was 1.45 percentage points. Furthermore, the Simple CNN model also improved the recall value of the Phoma label from 0.87 to 0.92. This was an improvement of 0.05, which indicated that Simple CNN identified fewer false-negative Phoma instances than MobileNetV2. This was relevant since disease identification aimed to minimize false negatives.

However, Simple CNN was not automatically the superior model. The Simple CNN required approximately 82 minutes for training, compared to around 16 minutes for MobileNetV2. This was because only 3,843 parameters were trainable in the main run of MobileNetV2, compared to 1.44 million for the Simple CNN model. Training time was an important criterion when quick model testing was crucial, thus making MobileNetV2 more efficient.

The confusion matrix results revealed that both models could effectively classify healthy leaves, as seen by their high values in the True Positive and True Negative cells. However, MobileNetV2 could identify all healthy test samples but classified 18 images labeled as Phoma into the leaf rust category. In contrast, the Simple CNN made mistakes when identifying healthy samples but minimized Phoma-to-rust confusion by classifying 12 Phoma samples as leaf rust instead. Thus, Simple CNN seemed to better distinguish the two classes in this run.

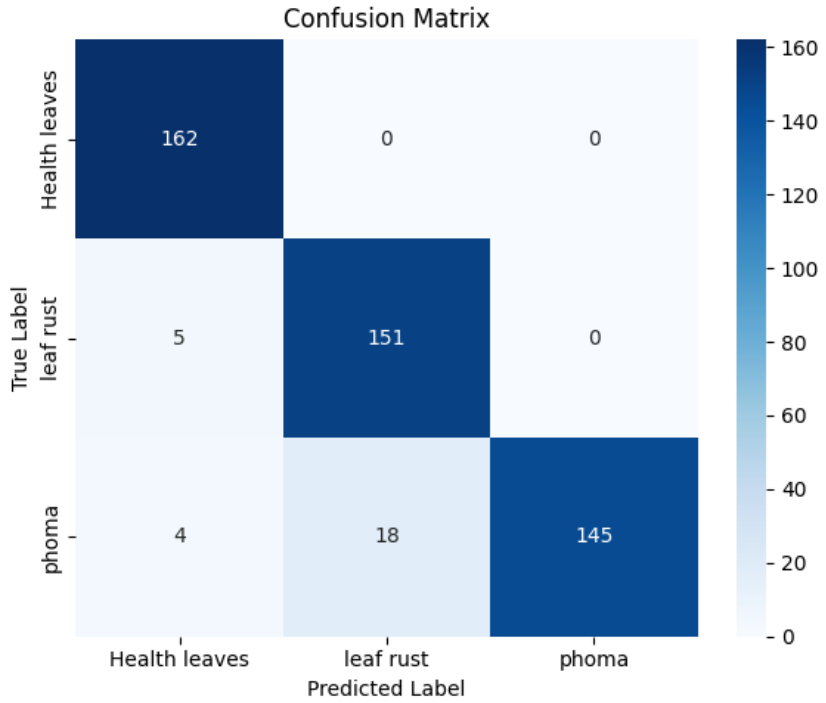


Fig. 4. Confusion matrix of MobileNetV2 on the test set.

However, besides the absolute number of accurate predictions, an important aspect of comparing model accuracy lies in the kind of errors each of them produces. MobileNetV2 classified all healthy test images accurately; however, it misclassified 18 Phoma samples as leaf rust. On the contrary, Simple CNN lowered the rate of Phoma misclassification to 12, but made a couple of misclassifications for healthy images. This suggests that classification of healthy leaves seems to be relatively simpler than Phoma and leaf rust recognition because the latter categories are too visually similar to one another. Such results also agree with common practice in the area of coffee disease recognition, which frequently makes mistakes on samples having visually similar patterns such as spots and lesions appearing on the leaf surface [9].

The difference in the performance of the models equals 1.45% or, in other words, no more than seven images out of 485 tested samples. Therefore, these numbers cannot be interpreted in favor of Simple CNN as being superior to MobileNetV2 in terms of accuracy. Instead, a fair comparison should include repeating the training process multiple times, taking the mean and standard deviation of accuracies and macro F1-scores for all iterations, and ideally including the statistical evaluation of paired sample predictions.

Another reason that can explain the low accuracy obtained by MobileNetV2 could be the use of a non-optimal pre-processing step before training. As stated in the Keras documentation, mobileNetV2.preprocess_input must be employed, meaning that the image pixels' values need to be scaled to [-1, 1]. Moreover, during the process of training MobileNetV2, most of its layers remained unchanged, meaning that the network did not fully adjust to the characteristics of the coffee leaf samples.

Keras confirms the MobileNetV2 preprocessing requirement, and its transfer-learning guide says fine-tuning can improve results but can also overfit if not done carefully [15].

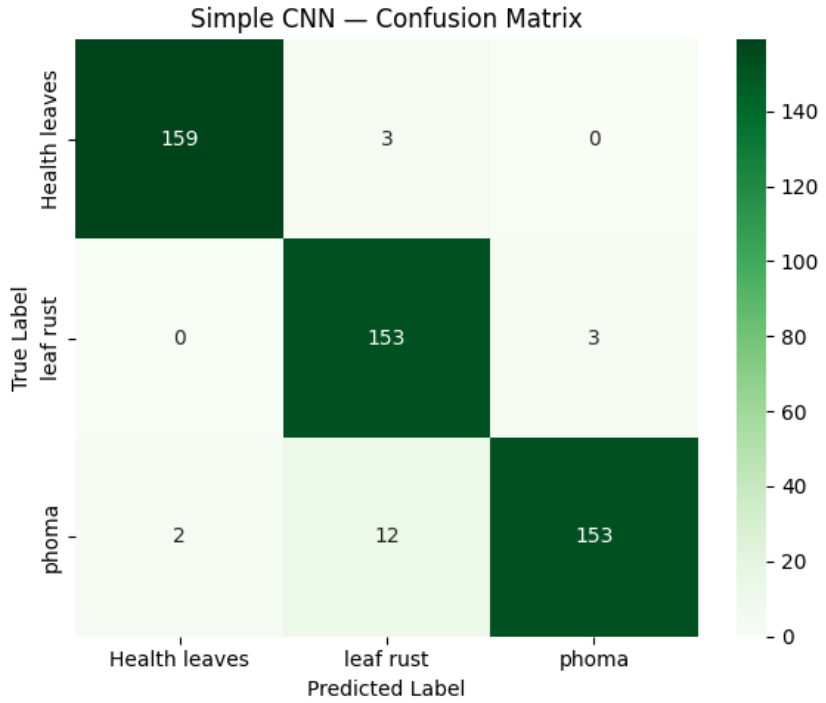


Fig. 5. Confusion matrix of Simple CNN on the test set.

Finally, care was needed in interpreting the result since the dataset source mentioned that augmentation techniques were utilized to improve sample diversity [1]. Augmented versions of the same leaves might have been included in each split, leading to an overestimated measurement of accuracy. While random training, validation, and test folders were generated, there was no evidence that the data were split in a manner that ensured each sample group contained similar augmented variants.

9. Limitations and Improvements

The biggest limitations of this experiment are methodological limitations rather than simply those pertaining to the choice of the model:

1. MobileNetV2 was not preprocessed optimally. In particular, image rescaling to [0, 1] range instead of using the standard Keras MobileNetV2 preprocessing function was done.

2. Fine-tuning of MobileNetV2 was not tested on the test set. As a consequence, it cannot be concluded definitively that the frozen base version performed slightly better than the Simple CNN on the test set in this particular run.

3. Only one train/validation/test split and one random seed were employed. With an accuracy difference of seven images only, repeating this experiment several times is required in order to come to a conclusion.

4. Possible presence of duplicates or augmented images was not considered fully. Should some images of the same original leaf appear in each of these three sets, the accuracy of the experiment would be biased.

5. Neither validation nor any external testing was performed. The ability of this model to recognize diseased leaves on the data from RoCoLe or JMuBEN/JMuBEN2 datasets must be tested to determine whether the model generalizes outside of the local dataset.

6. Explainability method was omitted. Application of Grad-CAM is required to ascertain whether the models actually pay attention to the actual signs of disease and not other factors such as leaf boundaries, backgrounds, or lighting [14].

Adding more models and comparing the accuracies is not necessary, since it is important to make this experiment truly scientifically valid. The suggested next experiment is as follows: fix MobileNetV2 preprocessing, test frozen and fine-tuned versions separately, run both models at least five times using different random seeds, present means and standard deviations for these values, perform perceptual hashing to detect duplicates, and provide some samples of Grad-CAM output for classification mistakes.

This problem is known as data leakage. Data leakage leads to overly optimistic results [13]. Class imbalance negatively impacts CNN's effectiveness in some cases [12].

10. Conclusion

This report provided a comparison of two models on a specific dataset containing coffee leaf disease images. The Simple CNN model gave the best test performance results with 95.88% accuracy. MobileNetV2 achieved the second-best test accuracy, with 94.43%. Moreover, MobileNetV2 also showed high F1-scores. It is worth mentioning that the MobileNetV2 model was trained much faster as only the classifier was trained on the pre-trained base model. Therefore, the result did not show that one model was always the best, but that each model had its own advantages. The Simple CNN model gave better results in this notebook run, but MobileNetV2 was more efficient during training.

Both models were used for the comparison. On the other hand, some limitations were also taken into account. Thus, among the technical errors, there were the use of incorrect MobileNetV2 preprocessing, lack of evaluation of the fine-tuned MobileNetV2 model separately, and possibility of data leakage due to image augmentations.

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